

DATA WAREHOUSING AND DATA MINING			
Course Code	25MCA204	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Prerequisites	Advanced Database Systems Mathematical Foundation for Computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Provide the student with an understanding of the concepts of data warehousing, its architecture and modelling		
2.	Understand the Knowledge Discovery Process and get the notion of Data Mining		
3.	Learn the Preprocessing techniques required for Data Mining		
4.	Learn the data mining tasks of Association Analysis, Classification and Clustering and also study their well-known techniques		
5.	Test real data sets using popular data mining tools		
UNIT – I			
Data Warehousing & modelling:			10 Hours
Basic Concepts: Data Warehousing: A multitier Architecture, Data warehouse models: Enterprise warehouse, Data mart and virtual warehouse, Extraction, Transformation and loading, Data Cube: A multidimensional data model, Stars, Snowflakes and Fact constellations: Schemas for multidimensional Data models, Dimensions: The role of concept Hierarchies, Measures: Their Categorization and computation, Typical OLAP Operations			
UNIT – II			
Introduction to Data mining:			10 Hours
Introduction, What is data mining, Motivating Challenges, The Origins of Data Mining, Data Mining Tasks Data: Types of Data - Attributes and Measurements, types of Data Sets – Record data, graph based data, ordered data, Data Quality, Data Preprocessing – aggregation, sampling, dimensionality reduction, feature subset selection, feature creation, discretization and binarization, variable transformation, Measures of Similarity and Dissimilarity – Euclidean distance measures, similarity measures for binary data, simple matching coefficient, Jaccard coefficient and cosine similarity			
UNIT – III			
Association Analysis:			10 Hours
Problem Definition, Frequent Item set Generation, Rule generation. Alternative Methods for Generating Frequent Item sets, FP Growth Algorithm, Subjective and Objective measures for Evaluating Association Patterns			
UNIT – IV			
Classification:			10 Hours
Preliminaries, General Approach to solving classification Problem, Decision Tree Induction, Rule Based Classifiers, Nearest Neighbor Classifiers, Bayesian Classifiers, Evaluating the performance of classifier.			

UNIT – V											
Clustering Analysis:										10 Hours	
Overview, K-Means, Agglomerative Hierarchical Clustering, DBSCAN, Density-based Clustering, Graph-based clustering, Cluster Evaluation- Classification oriented measures, Similarity oriented measures											
Course Outcomes: At the end of the course student will be able to											
1.	Articulate the architecture of Data Warehouses and compare design schemas										
2.	Apply various preprocessing tasks and define Data Mining										
3.	Understand Association rule mining and apply association algorithms										
4.	Learn to apply various classification techniques and algorithms and evaluate										
5.	Interpret and Understand various clustering techniques and algorithms										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA204.1		3	3	3	3	3	3	1	1	2	3
25MCA204.2		3	3	3	3	1	1	1	3	2	3
25MCA204.3		3	3	3	3	3	1	1	3	2	3
25MCA204.4		3	3	3	3	3	1	1	3	2	3
25MCA204.5		3	3	3	3	3	1	1	3	2	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson, Fourth impression, 2018.										
2.	Jiawei Han, Micheline Kamber, Jian Pei: Data Mining -Concepts and Techniques, 3 rd Edition, Morgan Kaufmann Publisher										
REFERENCE BOOKS:											
1.	Arun K. Pujari : Data Mining Techniques, 3 rd Edition, Universities Press, 2013.										
2.	Sam Anahory, Dennis Murray: Data Warehousing in the Real World, Pearson, 10 th Impression, 2012.										
3.	Michael J. Berry, Gordon S. Linoff: Mastering Data Mining, Wiley Edition, 3 rd Edition, 2012										
E Books / MOOCs/ NPTEL											
1.	NPTEL Course on Data Mining, IIT Kharagpur, 2024										